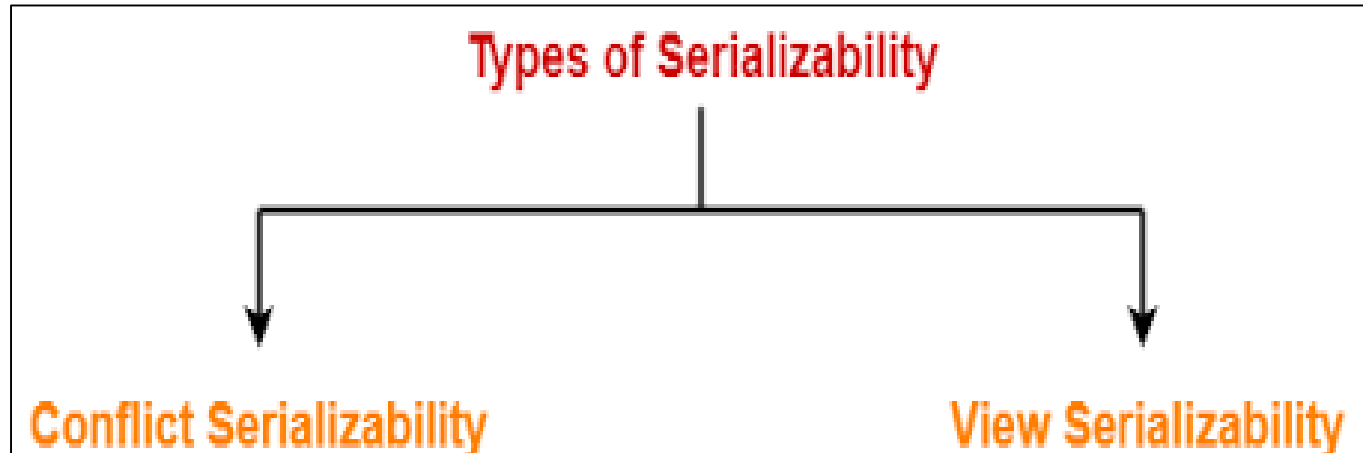


# Database Management System (BCSC – 1003)

## Topic: **DBMS Serializability**



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# Topics to be covered



- Serializability
- Serializability Schedule
- Types of Serializability
- Recoverability of Schedule

# Serializability

- When multiple transactions are running concurrently then there is a possibility that the database may be left in an inconsistent state.
- Serializability is a concept that helps us to check which schedules are serializable.
- A serializable schedule is the one that always leaves the database in consistent state.

# Serializable Schedule

- A serializable schedule always leaves the database in consistent state.
- A serial schedule is always a serializable schedule because in serial schedule, a transaction only starts when the other transaction finished execution.
- However a non-serial schedule needs to be checked for Serializability.
- A serial schedule doesn't allow concurrency, only one transaction executes at a time and the other starts when the already running transaction finished.

# Types of Serializability



There are two types of Serializability.

1. Conflict Serializability &

2. View Serializability

# Conflict Serializability

- A schedule is called conflict serializable if we can convert it into a serial schedule after swapping its non-conflicting operations.

## *Conflicting operations:*

- Two operations are said to be in conflict, if they satisfy all the following three conditions:
  1. Both the operations should belong to different transactions.
  2. Both the operations are working on same data item.
  3. At least one of the operation is a write operation.

# Conflict Serializability

Lets see some examples to understand this:

Exp 1: Operation  $W(X)$  of transaction  $T1$  and operation  $R(X)$  of transaction  $T2$  are conflicting operations, because they satisfy all the three conditions mentioned above. They belong to different transactions, they are working on same data item  $X$ , one of the operation in write operation.

Exp 2: Operations  $W(X)$  of  $T1$  &  $W(X)$  of  $T2$  are conflicting operations.

Exp 3: Operations  $W(X)$  of  $T1$  and  $W(Y)$  of  $T2$  are non-conflicting operations because both the write operations are not working on same data item so these operations don't satisfy the second condition.

Exp 4:  $R(X)$  of  $T1$  and  $R(X)$  of  $T2$  are non-conflicting operations because none of them is write operation.

Exp 5:  $W(X)$  of  $T1$  and  $R(X)$  of  $T1$  are non-conflicting operations because both the operations belong to same transaction  $T1$ .

# View Serializability

- View Serializability is a process to find out that a given schedule is view serializable or not.
- To check whether a given schedule is view serializable, we need to check whether the given schedule is View Equivalent to its serial schedule.
- A schedule will view serializable if it is view equivalent to a serial schedule.



# View Serializability

## *View Equivalent*

- Two schedules S1 and S2 are said to be view equivalent if they satisfy the following conditions:
  1. Initial Read: An initial read of both schedules must be the same. Suppose two schedule S1 and S2. In schedule S1, if a transaction T1 is reading the data item A, then in S2, transaction T1 should also read A.
  2. Update Read: If in schedule S1, the transaction T1 is reading a data item updated by T2 then in schedule S2, T1 should read the value after the write operation of T2 on same data item.
  3. Final write: A final write must be the same between both the schedules. In schedule S1, if a transaction T1 updates A at last then in S2, final writes operations should also be done by T1.

# View Serializability

*For Example:*

| T1      | T2       |
|---------|----------|
| Read(A) | Write(A) |

**Schedule S1**

| T1      | T2       |
|---------|----------|
| Read(A) | Write(A) |

**Schedule S2**

Above two schedules are view equivalent because Initial read operation in S1 is done by T1 and in S2 it is also done by T1.

# Recoverability of Schedule

- Sometimes a transaction may not execute completely due to a software issue, system crash or hardware failure.
- In that case, the failed transaction has to be rollback.
- But some other transaction may also have used value produced by the failed transaction.
- So we also have to rollback those transactions.

# References



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*Thank  
you*

